P510/1
PHYSICS
Paper 1
(Theory)
August 2026
3 hours

MASAKA DIOCESAN EXAMINATIONS BOARD
JOINT MOCK EXAMINATIONS 2026
Uganda Advanced Certificate of Education
PHYSICS
Paper 1
(Theory)
3 hours

INSTRUCTIONS TO CANDIDATES:

*This examination paper consists of **two** sections; **A** and **B**. It has **six** items.*

*Section **A** has **two** compulsory items.*

*Section **B** has **two** parts; **I** and **II**. Respond to only **one** item from each part.*

*Respond to **four** items in all.*

*Any additional item(s) responded to, will **not** be scored.*

Responses to each item should start on a fresh page.

Silent non-programmable scientific calculators may be used.

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SECTION A

*Respond to **all** the items in this section*

Item 1

While at the airport, a pilot conducts a final check on whether his aircraft with a total wing area of 40 m^2 can ably create sufficient lift to takeoff. Takeoff occurs when the lift exceed it's the total weight of the aircraft. Flight instruments indicate that air speed above the wings is 100 ms^{-1} while that below is 120 ms^{-1}. He does not know how to interpret the measurements in relation to the shape of the aircraft wings to create a lift force. He also wants to know whether these speeds are sufficient to cause lift.

During the flight, the aircraft's communication and navigation system fails, and the pilot carries out an emergency evacuation of the passengers using a parachute. Just 2 m above the ground Musa's parachute developed a puncture and felt weightless as he descended. Upon reaching the ground, he flexed his knees while the other passengers reported of normal weight and did not flex their knees. Musa and those passengers narrated to situation to the rescue team who were waiting. The team did not understand this.

Dr. Sarah is tasked with deploying a 2000 kg emergency communication satellite to locate the wreckage of the aircraft and restore regional links. She aims for a stable orbit at a height of 800 km above the earth's surface. She has been advised that the mission will succeed only if the launch energy needed stays within their rocket's capacity of 6.5×10^{10} J. She needs TO evaluate the success of the mission, be guided on the launching procedures, understand what is likely to happen in case the propulsion system fails when it has just reached this altitude in order to be able to estimate when it likely to return to the earth's surface.

Hint: Density of air $\rho_a = 1.29\ gm^{-3}$

Mass of Musa, 70 kg

Total weight of the aircraft = 15,000 kg

Acceleration due to gravity at the earth's surface, $g = 9.81 ms^{-2}$.

Radius of the earth, $M_e = 6.4 \times 10^6$ m.

Gravitational constant, $G = 6.67 \times 10^{-11} Nm^2 kg^{-1}$

Tasks

(a) Respond to the concerns of the pilot and the rescue team.

(b) Help Dr. Sarah to accomplish her launching mission.

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Item 2

An industrial manufacturing plant utilizes a modern diagnostic X-ray tube operating at 40 kV to maintain standard thickness gauge of aluminium sheets produced by a certain factory. The automated gauge uses a radioactive source that emits beta particles. The existing hydroelectric power grid faces a daily deficit of 7.5×10^{13} J to run the factory. Therefore, a small-scale nuclear fission reactor has been set up to cater for the deficit.

The workers in the factory do not know how nuclear fission releases energy, how such systems are regulated to maintain safety and why a neutron is used not any other particle such as alpha or a proton. In an effort to educate the workers, the engineer has provided a standard nuclear equation for the nuclear process:

$$^{235}_{92}U + {}^{1}_{0}n \longrightarrow {}^{141}_{56}Ba + {}^{92}_{36}Kr + 3{}^{1}_{0}n$$

Due to limited availability of uranium, the workers want to know whether 1 kg of pure uranium is enough to produce the daily deficit energy.

While meeting a group of physics learners on study trip at the factory, the quality assurance team rely on the atomic structure of aluminium which is known to possess an interplaner atomic spacing of 2.3×10^{-10} m. The students are further informed that the aluminium is flaw free diffracting angle for X-rays does not fall below 7° for the second order diffraction if the team assigns the students with verifying the standards of produced aluminium sample sheets, provide detailed procedure of how the factory maintains uniform thickness of a given aluminium sheet and how the detector identifies the radiations involved.

The team is left with the task of advising the factory workers why a beta emitting source is preferred to any other radiation.

Hint:

Mass of $^{235}_{92}U = 235.0439u$, Mass of $^{1}_{0}n = 1.0087u$, Mass of $^{141}_{56}Ba = 140.9144u$, Mass of $^{92}_{36}Kr = 91.9262u$, $1u = 931MeV$

Speed of electromagnetic radiation in vacuum, $c = 3.0 \times 10^{8} ms^{-1}$

Plunk constant, $h = 6.6 \times 10^{-34} Js$.

Electronic charge, $e = 1.6 \times 10^{-19} C$.

Tasks:

(a) Analyse the information and educate the factory workers about their concerns.

(b) Help the students and the quality assurance team to carry out its obligations.

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SECTION B
PART I

Item 3

A newly established agricultural research center in your community is developing solar-powered crop dryers to minimize post-harvest losses. The main components of the solar system are triangular prism of refracting angle, 60° to isolate infrared rays from harmful ultraviolet rays that degrade nutrients. An array of six identical rectangular solar plates measuring 2.0 m by 1.5 m as a perfect black body to absorb the solar energy that run the crop dryers. The team sets up the system in a way that sunlight strikes the first face of the glass prism at an angle of incidence 45° and infrared passes symmetrically through the glass prism. A triangular glass prism can effectively isolate these rays if their angular separation exceeds 0.46°. The team is interested in evaluating how the incoming sunlight behaves upon entering the glass prism to isolate heating rays from the harmful ones and whether it can serve the intended purpose.

It has come to the notice of the station manager that the six solar plates absorb energy from the sun at a standard solar constant of1350Wm^{-2}. The manager wants to monitor these extreme temperatures generated at the focal spot of the dryer without melting standard contact sensors. The team has proposed to the manager to choose between bolometer strip and an optical pyrometer. However, the manager doesn't know the basis of selection of the suitable measuring device. He also doesn't know how the selected device can be used to monitor the temperatures.

Hint: Refractive index of glass prism for ultraviolet rays, $n_{uv} = 1.525$.
Stefan's constant, $\sigma = 5.67 \times 10^{-8}\text{Wm}^{-2}\text{K}^{-4}$.

Tasks:

(a) Analyse the information and address the concerns of the engineering team.

(b) With justification, help the manager to monitor the operating temperatures.

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Item 4

A community resource center is preparing to host an interactive physics seminar for high schools in the area. The seminar relies on diesel generator managed by a technician to power both public address and a digital projector setup by the facilitator.

During a test run of the generator engine, the technician observes that 1 mole of oxygen at s.t.p behaves ideally inside the engine cylinder when undergoing an extreme rapid stroke. The gas is swiftly compressed to a volume 0.0166 m^3 before any heat can escape to the surroundings. This sudden compression requires the piston to perform 350 J of mechanical work on the gas, rising its temperature to 290 K. The technician does not know why the rising and is worried that the resulting internal pressure might exceed the engine block's structural safety limit of 1.64×10^5 Pa, causing an explosion that may disrupt the seminar's power supply. He needs to examine the thermodynamic behavior.

Meanwhile the facilitator sets up the projector using a single standard convex lens of focal length 12 cm, positioning the slide exactly 10 m from the projection screen. When she projects the first sample slide which displays an experimental data plot a plane progressive wave for the student's seminar assignment, the text and equations appear completely unreadable. The learners complain that the image is badly blurred, lacks sharp focus at the edges, and suffers from distracting rainbow- coloured fringes around the text characters. The facilitator needs to analyze the optical specifications needed to modify the lens setup for a perfect focus of the image. The progressive wave equation on the blurry slide is intended to show: $y = 0.1 \sin\left(200\pi t - \frac{20\pi}{17}x\right)$, where y is in mm, t is in seconds and x is the distance from a fixed origin in metres.

Without correcting these optical defects, the seminar cannot proceed, and the learners will be unable to analyze the wave's speed or deduce the modified equation for a secondary wave that has twice the amplitude, double the frequency and travels in the exact opposite direction. Finally as part of the seminar, the learners want to understand how the modified projector works to produce a clear display.

Hint:

Standard atmospheric pressure $= 1.01 \times 10^5$ Pa.
Density of oxygen at s.t.p $= 1.43$ kgm^{-3}
Molar mass of oxygen $= 32g$

Tasks

(a) Analyze the thermodynamics process to address technician's safety concerns to evaluate if the engine is at risk of an explosion.

(b) Help the facilitator with an illustration to diagnose and fix the optical defects causing the poor display, and provide the solutions to the assignment for the students once the display is clear.

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PART II

Item 5

A smart car garage in a certain city is hosting a group of mechanic trainees. One day, they have received a customer complaining of his car's cabin bulb and electric motor which were not responding properly. The mechanic trainees suspected that the fault lay within the wiring system displayed in figure 1. M is a device that runs a cooling fan.

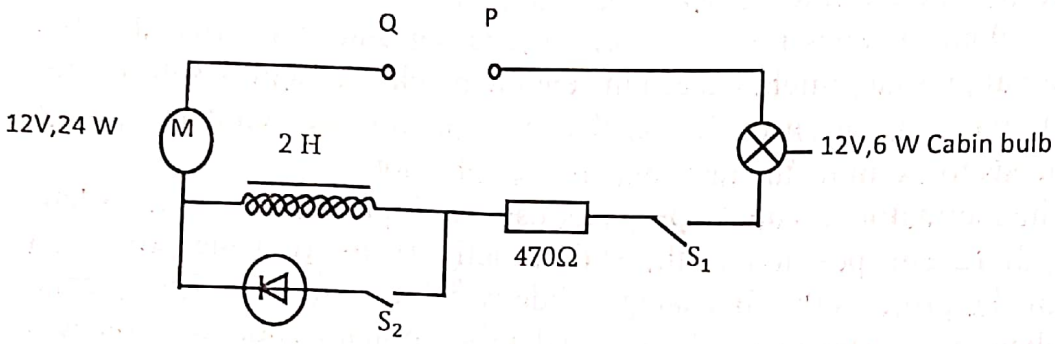

Fig. 1

As part of their tests they connected terminals P and Q across an a.c generator operating at 100 V (r.m.s), 50 Hz. When they closed switch, S_1, both the cabin bulb and the cooling fan could not respond. When all the switches, were closed only the cabin bulb responded while the cooling fan completely failed to run. One of the mechanic trainees, Tom claimed the response failure was due to blow off of the coil since the potential difference across it had exceeded 8.0 V. They need to verify whether Tom's claim is valid and analyse the behavior of the circuit components under these conditions. Another member, Sarah suggested that replacing the a.c supply with a d.c battery supply with the same rating while keeping terminal P positive with respect to maintaining switch S_2 open could be a better option for all the components to respond properly. However, she has feared that the cabin bulb may blow, she wants to know the value of a shunt which could be connected across the bulb to prevent from the blow.

Tasks:

(a) Analyse the information and address the concerns of the mechanic trainees to evaluate the performance of the first power source.

(b) Use the details provided to address the requirements of Sarah's second power source, and describe the operation of the device that runs the cooling fan.

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Item 6

Testing facilities use sensitive meters to measure electricity which are very delicate to handle. A technician is assigned to evaluate the sensitivity of a moving coil galvanometer with an internal resistance 2 Ω. The instrument consists of a rectangular coil with 500 turns each of area $4.0 \times 10^{-4}\text{m}^2$ suspended in a radial magnetic field of flux density 0.25 T by hairsprings with torsional constant, $3.0 \times 10^{-5}\text{Nmrad}^{-1}$.

To examine its response limits, the galvanometer is connected in series with a 0.5 m bare copper wire as a protective resistor. The technician repeatedly charges an air capacitor to 30 V and then discharges it through the galvanometer using a vibrating switch working at 50 times per second. When the circuit is activated, the galvanometer deflects steadily through 0.4 radians. The technician is interested in knowing the energy changes that account for deflection of the galvanometer. He does not know how the wire will eventually heat up under operation. He worries that if the heat generated in the constantan wire reaches 1.62×10^{-5} J, in in a span of half a minute after the switch has been activated, the delicate hairsprings may burn out.

Following the initial test, the technician attempts to re-use the components to assemble a radio receiver tuning circuit designed to receive incoming radio signals at exactly 2.516 MHz. He disconnects the wire from the previous circuit and uses the wire to wind a compact solenoid of length 22.7 cm with 600 turns and a cross-sectional radius 1.42 cm. He connects this new inductor in series with the capacitor and the galvanometer across a 12 V(r.m.s), 40 Hz a.c source.

He wants to measure the current through the circuit, but when he switches on the source, the galvanometer needle seemed not to deflect. He is unable to take readings and does not know why. He wants assistance to understand how the tuning circuit can be modified so that the galvanometer can register a readable current.

Hint:

Cross sectional area of copper wire, $A = 4.25 \times 10^{-10}\text{m}^2$

Electrical resistivity of constantan wire, $\rho = 1.7 \times 10^{-8}\Omega\text{m}$

Permeability of free space, $\mu_o = 4\pi \times 10^{-7}\text{Hm}^{-1}$

Tasks:

(a) Address the concerns of the technician to protect the galvanometer.

(b) Help the technician to use the galvanometer record current through the designed circuit, estimate the reading and verify whether it will be able to detect the incoming radio signals.

END

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Item one

$A = 40m^2$

- Speed above wings, $V_1 = 100 m/s$.
- Speed below wings, $V_2 = 120 m/s$.
- $\rho_{air} = 1.29 kg m^{-3}$.
- Aircraft mass, $m = 1500kg$.
- $g = 9.81 ms^{-2}$.

a) Using Bernoulli's Principle

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2.$$

$$P_1 - P_2 = \frac{1}{2}\rho(v_2^2 - v_1^2)$$

$$= \frac{1}{2} \times 1.29 (120^2 - 100^2)$$

$$= 0.645(14400 - 10000)$$

$$= 2838 Pa.$$

$$F = PA.$$

$$= 2838 \times 40$$

$$= 1.1352 \times 10^5 N$$

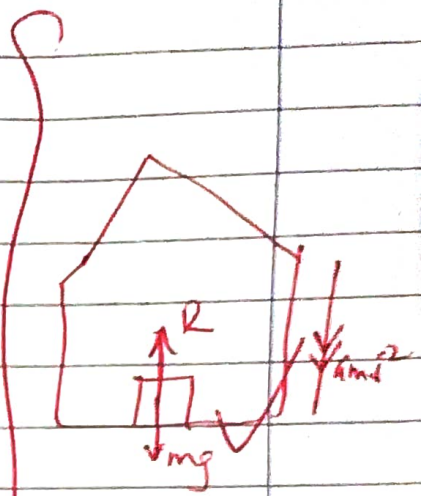
Weight of aircraft.

$$W = mg.$$

$$= 15000 \times 9.81$$

$$= 147150$$

$$= 1.4715 \times 10^5 N$$



$m = 70kg$

$F = ma.$

$mg - R = ma.$

$R = mg - ma$

$= (70 \times 9.81 - 70 \times 9.81)$

$R = 0N$

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flex - "soft landing"
not flexing - "hard landing"

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Since W > F, lift force is not sufficient for take off.

b) → Explanation of take off using Bernoulli's principle.
- An aircraft has a propelor which makes air molecules above wings with higher velocity, creating low pressure zone. Air molecules below wings have lower velocity, hence a higher pressure zone. Difference in pressure creates a net force called lift force necessary for take off.

c) → Musa felt weightless after the parachute developed a puncture because he entered free fall.
- During free fall $a = g$, the only force acting on him was his weight. His apparent weight, $R = 0$.
- Therefore, he experienced weightlessness.

d) → Musa flexed his knees, for him to have a soft landing ($Ft = I$). This increases the time of impact on the ground (impulse) and therefore reduces injuries.
- other passengers who never flexed had hard landing. Therefore time of impact was small and more pain was felt.

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$m = 2000kg$

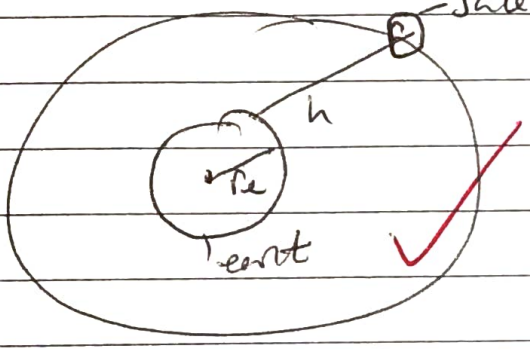
$R_e = 6.4 \times 10^6 m$

$h = 800km = 8.0 \times 10^5 m$

$R = r_e + h$

$= 6.4 \times 10^6 + 8.0 \times 10^5$

$= 7.2 \times 10^6 m$



Centripetal force = Gravitational force

$m_s w^2 R = \frac{G m_s m_e}{R^2}$

$w^2 R = \frac{G m_e}{R^2}$

$G m_e = w^2 R^3$

but $g R_e^2 = G m_e$

$g R_e^2 = G m_e$

$m_e = \frac{g R_e^2}{G}$

$= \frac{9.81 \times (6.4 \times 10^6)^2}{6.67 \times 10^{-11}}$

$= 6.02 \times 10^{24} kg$

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– Energy required to place satellite in orbit
$(E = P.E + K.E)$

$= GMm\left(\frac{1}{R} - \frac{1}{2R}\right)$

$= GM_e m_s\left(\frac{1}{R} - \frac{1}{2R}\right) = -\frac{GM_e m_s}{2R}$

$= 6.67\times10^{-11} \times 6.02\times10^{24} \times 2000 \left(\frac{1}{6.4\times10^{6}} - \frac{1}{2\times7.2\times10^{6}}\right)$

$= 5.5769\times10^{10}\ J < 6.5\times10^{10}\ J$

Since $6.98\times10^{10} > 6.5\times10^{10}\ J$

–

→ Orbital speed at 600km

$\frac{mv^2}{R} = \frac{GM_s M_e}{R^2}$

$V = \sqrt{\frac{GM}{r}}$

$V = \sqrt{\frac{GM_e}{R}}$

$= \sqrt{\frac{6.67\times10^{-11} \times 6.02\times10^{24}}{7.2\times10^{6}}}$

$\simeq 7.47\times10^{3}\ ms^{-1}$

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launch Procedure;
- launch vertically initially ✓
- Gain sufficient altitude ✓
- Gradually tilt the rocket horizontally
- Increase horizontal velocity
- make orbit circular

} (02) any 2

→ If rocket has already achieved its orbital speed, it will continue orbiting Earth since gravity provides centripetal force.
- If propulsion fails before sufficient horizontal velocity achieved, it will fall back towards Earth.

- $h = \frac{1}{2} g t^2$

$t = \sqrt{\frac{2h}{g}}$

$= \sqrt{\frac{2 \times (8.0 \times 10^5)}{9.81}}$

=

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Item 2:

- Fission reaction leads to release of energy because from Einstein's mass-energy relation $\Delta E = \Delta m c^2$. Implying change in mass leads to change in energy. So mass of products is slightly less than mass of reactants. This difference in mass (mass defect) accounts for energy released.

- mass of uranium, $U = 235.04394$
- mass of neutron, $n = 1.00874$
- mass of Barium, $Ba = 140.91444$
- Mass of krypton, $Kr = 91.92624$.

total mass of reactants;

$235.0439 + 1.0087 = 236.0526$.

total mass of products;

$140.9144 + 91.9262 + 3(1.0087)$

$= 235.8667$.

Mass defect, $\Delta m = (236.0526 - 235.8667)$

$= 0.1859$

Energy released;

Given

$1u = 931 MeV$

$E = (0.1859 \times 931)$

$\simeq 173.0729 MeV$.

Energy in joules

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1 MeV = $1 \times 10^6 \times 1.6 \times 10^{-19}$ J

173.0729 MeV = $(173.0729 \times 1 \times 10^6 \times 1.6 \times 10^{-19})$ J

= $2.7691664 \times 10^{-11}$ J ✓

— Energy from 1kg of uranium;

— Mass of one uranium atom = $235.0439 \times 1.66 \times 10^{-27}$

≃ 3.90×10^{-25} kg.

— Number of atoms in 1kg;

$N = \frac{1}{3.90 \times 10^{-25}}$

= 2.56×10^{24} ✓

— Total energy = Number of atoms in 1kg × Energy released during fission

= $(2.56 \times 10^{24} \times 2.7691664 \times 10^{-11})$

= 7.1×10^{13} J

Since $7.1 \times 10^{13} < 7.5 \times 10^{13}$ J, 1kg of pure uranium is not enough to generate electricity needed.

How such systems (nuclear reactor) is controlled;

- Control rods: absorb excess neutrons
- Moderator: slows neutrons to increase the fission
- Coolant: removes heat.
- Shielding: absorbs harmful radiations
- Containment structure: prevents radioactive materials from escaping
- Automatic shut down system controls the chain reaction

any (3)

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Waste ... is buried in lead-...
Chernobyl - Ukraine - 01:23 - 26/4/1986 - reactor No 4 → 31 People died, 7 wks
Holland - animals destroyed

State news agency (TASS) New York Times

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— A neutron is used since it has no charge. Therefore it does not experience electrostatic repulsion from the positively charged uranium nucleus, so it can easily enter and cause fission.

— A proton or Alpha particle is positively charged and would experience strong electrostatic repulsion.

b). Tube Voltage

$V = 40\,KV = 40{,}000\,V$

K.e, $E = eV$

$= (1.6 \times 10^{-19} \times 40{,}000)$

$= 6.4 \times 10^{-15}\,J$

$$E = \frac{hc}{\lambda}$$

$$\lambda = \frac{hc}{E}$$

$$= \left(\frac{6.6 \times 10^{-34} \times 3.0 \times 10^{8}}{6.4 \times 10^{-15}}\right)$$

$\lambda = 3.09 \times 10^{-11}\,m$.

$d = 2.3 \times 10^{-10}\,m$

using Bragg's law:

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$2d\sin\theta = n\lambda$

$n = 2$.

$2 \times 2.3 \times 10^{-10} \sin\theta = 2 \times 3.09 \times 10^{-11}$

$\sin\theta = 0.1345$

$\theta = 7.73^\circ$

required minimum angle is 7°

Since $7.73^\circ > 7^\circ$, the Aluminium sample satisfies the diffraction requirement.

→ How X-rays maintain uniform thickness;

- X-rays are directed through the Aluminium sheet.
- Some are absorbed by aluminium, others transmitted.
- Amount transmitted depends on the thickness.
- A detector measures the transmitted radiation.
- Detector signal is converted into thickness.
- If the sheet becomes too thick, less radiation reaches the detector; if thin more radiation reaches.
- An automatic control system can adjust the production process to maintain constant thickness.

→ Why beta radiation suitable;

- Beta have moderate penetrating power. Therefore they ~~have~~ are suitable for monitoring thickness such as thin metal sheets.

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Alpha - has very low penetration - so unsuitable
Gamma - has very high penetration - so may pass through thin material with less useful sensitivity

- Detector placed on opposite side measures beta radiation transmitted through the sheet.

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Item 3:

$A = 60°$

$i = 45°$

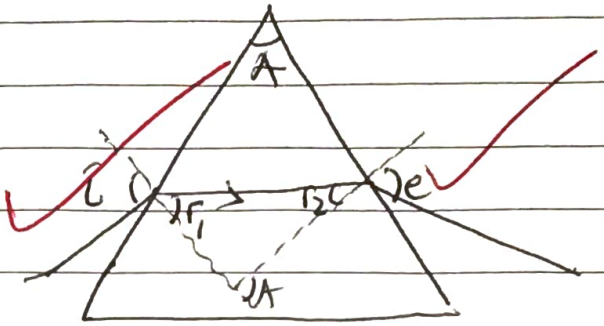
For infrared passing symmetrically:

$i = e = 45°$ ✓

also $r_1 + r_2 = A$

$r_1 = r_2$

$r = 30°$ ✓



Using Snell's law

$\frac{n_g}{n_a} = \frac{25}{25}$

$n_a \sin i = n_g \sin r_1$

$1 \times \sin 45 = n_{g1R} \cdot \sin 30$

$n_{g1R} = \frac{\sin 45}{\sin 30}$

$= 1.414$ ✓

For Ultra Violet radiation

$n_{uv} = 1.525$

$n_a \sin i = n_{guv} \sin r$

$\sin 45 = 1.525 \sin r$

$r = \sin^{-1}\left(\frac{\sin 45}{1.525}\right)$

$= 27.62°$ ✓

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at second face:

$r_2 = 60 - 27.62$

$r_2 = 32.38°$ ✓

Apply Snell's law:

$n_g \sin r_2 = n_a \sin e$

$1.525 \sin 32.38 = \sin e$ ✓

$e = \sin^{-1}(1.525 \sin 32.38)$

$= 54.74°$ ✓

Deviation of UV

$d = i + e - A$ ✓

$= 45 + 54.74 - 60$

$= 39.74°$ ✓

Deviation for Infrared:

$d = i + e - A$ ✓

$= 45 + 45 - 60$

$= 30°$ ✓

Angular separation $= 39.74 - 30 = 9.74°$ ✓

required separation $= 0.46°$

$9.74° > 0.46°$, the prism can effectively separate Ultra violet and Infrared radiations.

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- 6 plates
- Area of each Plate = $2.0 \times 1.5 = 3m^2$
- total area of 6 plates = $6 \times 3 = 18m^2$

Solar power/constant = $1350 Wm^{-2}$.

Power absorbed;

$P = IA$

$= 1350 \times 18$

$= 24300W$ (Power recieved by the plates).

→ For a black body:

$P = \sigma A T^4$

$$T = \left(\frac{P}{\sigma A}\right)^{\frac{1}{4}}$$

$$= \left(\frac{24300}{5.67 \times 10^{-8} \times 18}\right)^{\frac{1}{4}}$$

$T = 393K$ (temperature generated at the focal spot of the dryer)..

b) The ~~opti~~ optical Pyrometer is better than a bolometer ship since it can measure relatively high temperatures ~~to~~ without physical contact.

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Operation:

- The objective lens is focussed onto the hot object and eye piece adjusted such that the filament is in the same plane with it and the objective.
- Radiation from the hot object is focussed by the eye piece and the Rheostat adjusted such that its intensity merges light produced by the filament. At this point both filament and hot object are at same temperature.
- This temperature is read off from the ammeter initially calibrated to read temperatures directly hence measuring temperature of hot object.

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Item 4:

a) - Initially, $m = 32g = 0.032kg$.
- Density of oxygen $= 1.431 kgm^{-3}$.
- $T_1 = 273K$.

- Final temperature, $T_2 = 290K$
- Final volume $V_2 = 0.0166m^3$.
- work done on the gas, $\Delta W = 350J$

Initial volume;

$40 \times \frac{1}{2} = \frac{20}{20}$

$$V = \frac{m}{\rho}$$

$$V_1 = \frac{0.032}{1.43}$$

$$V_1 = 0.0224m^3$$

- Since $V_1 > V_2$, the gas has been compressed.

- Temperature increases because its process is extremely rapid, very little heat escapes. Also the work done on the gas increases its internal energy, hence increasing its temperature.

$$\Delta Q = \Delta U + \Delta W$$

$$\Delta Q = 0$$

$$\Delta U = \Delta Q - \Delta W$$

$$\Delta U = 0 - \Delta W$$

- In the scenario, work is done on the gas, not by the gas
- Therefore work done by gas is negative

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- When work is done on the gas → ΔU ↑

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- When work is done by gas → ΔU ↓

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$\Delta W = -350J$ (since compression)

$\Delta U = -(-350) = +350J$ using Convention

$\Delta U = 350J$

Therefore work done on gas leads to increase in ΔU.

— ~~Final Pressure~~; Final Pressure;

Using; $P \cdot V = nRT$

$$P = \frac{nRT}{V}$$

$$P = \frac{1 \times 8.31 \times 290}{0.0166}$$

$$P = 1.45 \times 10^5 Pa$$

Safety limit: $P = 1.64 \times 10^5 Pa$.

Since $1.45 \times 10^5 Pa < 1.64 \times 10^5 Pa$, The pressure is below the safety limit, so the engine is safe from explosion.

b) Projector and wave.

$f = 12cm$

object - screen distance

$D = 10m = 1000cm$

for a projector:

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$u + v = 1000cm$

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$\frac{1}{f} = \frac{uv}{u+v}$

$\frac{1}{f} = \frac{u+v}{uv}$ $\quad u+v = 1000$

$\frac{1}{12} = \frac{1000}{uv}$

$uv = 12000$ ——— (1)

$u \simeq 987.85\text{cm}$

$v \simeq 12.15\text{ cm}$

∴ The lens should be about 12.15cm from the screen and the slide about 987.9cm from the lens.

$M = \frac{v}{u}$

$= \frac{12.15}{987.85}$

$|M| \simeq 0.0123$

Optical defects

(a) Spherical abberation:

- This is the distortion of image by either a lens or a mirror

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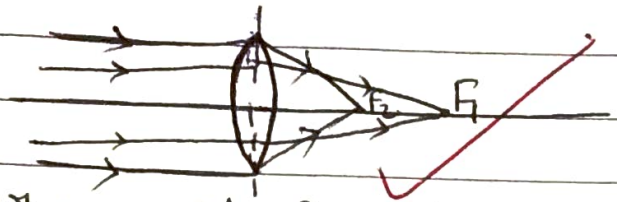
of wide aperture.



— When a wide of white light is incident on either a lens or a mirror of wide aperture, central rays converge at point far away from the the lens.

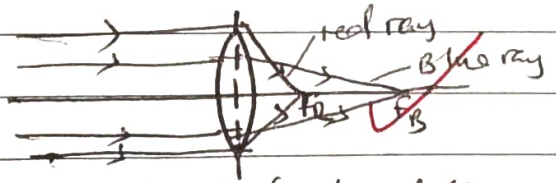
— Rays far from principal axis converge near the lens. Images formed are ~~formed~~ a series and blurred.

→ In lenses, spherical abberation is minimised by using a stopper ie opaque disc with central hole to cut off marginal rays, allows only paraxial rays.

— In mirrors, we use parabolic mirrors; since they converge wide beam of light incident on its surface to a single focus.

(ii) Chromatic abberation;

— Is the colouring of images produced by a lens.



— When white light is incident on a lens, different colour components are refracted by different amounts. Images corresponding to different colours are formed in different positions along the principle axis.

→ It can be reduced by using an achromatic doublet. It consists of a convex lens combined with a concave lens

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made from different glass materials. Convex deviates rays and concave nullifies the diversion.

Wave equation;

$y = 0.1\sin(200\pi t - x)$

Compare with;

$y = A\sin(\omega t - Kx)$

$A = 0.1mm$ ✓

$\omega = 200\pi$

$K = 1rad/m$ ✓

- frequency;

$\omega = 2\pi f$

$200\pi = 2\pi f$

$f = 100Hz$ ✓

- wavelength;

$K = \frac{2\pi}{\lambda}$

$1 = \frac{2\pi}{\lambda}$

$\lambda = 2\pi m$ ✓

- Wave velocity / speed.

$V = f\lambda$ ✓

$= 100 \times 2\pi$

$= 200\pi$ ~~$2.628 m/s$~~.

Modified wave;

twice amplitude. $A_2 = 0.2mm$.

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Double frequency; $f_2 = 200\,Hz$

Therefore $w_2 = 2\pi(200) = 400\pi$

If the wave travels in opposite direction then;

$K_2 = 2\,rad/m$.

$y = 0.2\sin(400\pi t + 2x)$, y is in mm.

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Item 5:

- Motor; 12V, 24W.
- Cabin bulb: 12V, 6W
- Inductor: 2H
- Resistor 470Ω.
- Ac supply: 100 $V_{r.m.s}$ 150Hz.

a) AC supply (Tom's Claim)

— Resistance of the motor;

$P = \frac{V^2}{R}$, $R = \frac{V^2}{P}$

$= \frac{12^2}{24}$

$= 6\Omega$

— Resistance of bulb:

$R = \frac{12^2}{6} = 24\Omega$

— Inductive reactance;

$X_L = 2\pi fL$

$= 2\pi \times 50 \times 2$

$= 628.3\Omega$.

— Total impedance

$R = 24 + 6 + 470 = 500\Omega$.

$Z = \sqrt{R^2 + X_L^2}$

$= \sqrt{(500)^2 + (628.3)^2}$

$= 803\Omega$.

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→ Circuit current;

$$I = \frac{V}{Z}$$

$$I = \frac{100}{803}$$

$$I \approx 0.125A \quad 0.1245A$$

→ Voltage across coil;

$$V_L = I X_L$$

$$V = 0.125 \times 628.3$$

$$V \simeq 78.2V$$

- So the coil voltage is much greater than 8V.

→ Why the motor fails;

Voltage across the motor

$$V = IR$$

$$= 0.125 \times 6$$

$$\simeq 0.747V < 12V$$

- Motor relieves far too little voltage, it cannot operate normally.

6) D.C Supply (Sarah's clam)

- If 100V DC source is used and switch S_2 remains open
- Inductor behaves like a short circuit
- $R_T = 470 + 6 + 24$

$$= 500\Omega$$

$$I = \frac{100}{500} = 0.20A$$

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- Shunt is low-resistance resistor connected in parallel with a device to provide an alternative path for current.

→ When D.C is switched on, inductor opposes I, when D.C flows for a long time current is constant $\frac{dI}{dt} = 0$, so no more voltage in inductor with A.C its continuous.

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Voltage across bulb!

$V = IR_b$

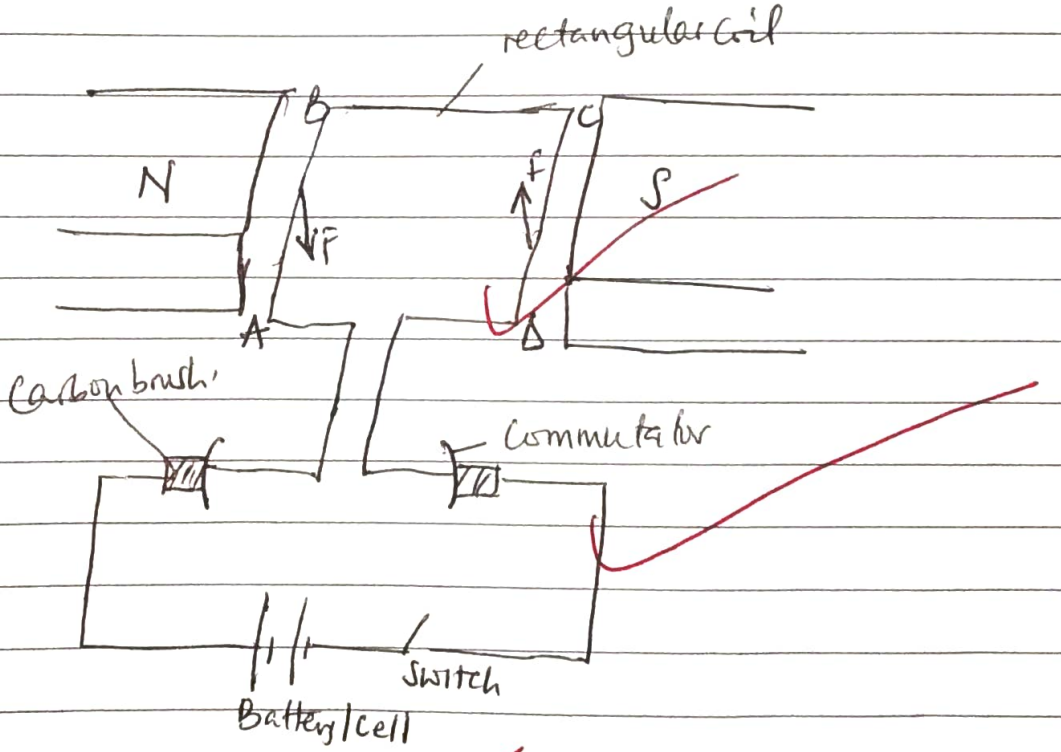
$= 0.2 \times 24$

$= 4.8V < 12V$, the rating of the bulb.

- bulb will not blow, it will be under powered and will produce dim lights. So a shunt is not necessary to protect it under these assumptions.

~~Operation of D.C motor used to run the fan:~~

- the device used to run the fan is a d.c motor.



- Current flows in the direction ABCD, when switch is closed. AB experiences downward force, CD upwards force, the two forces form a couple, as it rotates in anti clockwise.
- As the coil rotates, ~~direction~~ Commutator loose contact

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~~with from~~ with Carbon brushes and interchange positions
- direction of forces changes but direction of rotation remains constant.

ITEM 6:

i) Protecting galvanometer;

$N = 500$, $A = 4.0 \times 10^{-4} m^2$, $B = 0.25 T$, $K = 3.0 \times 10^{-5} Nm rad^{-1}$
$\theta = 0.4 rad$.

Torque on the coil;

$T = NBIA$

at equilibrium;

$NBIA = K\theta$

$$I = \frac{K\theta}{NBA} = \frac{3.0 \times 10^{-5} \times 0.4}{500 \times 0.25 \times 4.0 \times 10^{-4}}$$

$I = 2.4 \times 10^{-4} A$

$\simeq 0.24 mA$ (Current corresponding to 0.4 rad deflection).

- Resistance of the protective wire;

$\rho = 1.7 \times 10^{-8} \Omega m$, $L = 0.5 m$, $A = 4.25 \times 10^{-10} m^2$

$$R = \frac{\rho L}{A}$$

$$R = \frac{1.7 \times 10^{-8} \times 0.5}{4.25 \times 10^{-10}}$$

$R = 20 \Omega$

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- Power dissipated in the wire;

$P = I^2 R$

$= (2.4 \times 10^{-4})^2 \times 20$

$= 1.152 \times 10^{-6}$ W

$E = Pt$

$= (1.152 \times 10^{-6} \times 30)$

$= 3.456 \times 10^{-5}$ J

The limit is 1.62×10^{-5} J

Since $3.456 \times 10^{-5} > 1.62 \times 10^{-5}$ J; Protective wire would generate more heat than stated amount, therefore technician should not allow this operation.

- time to reach limit;

$t = E/P$

$= \dfrac{1.62 \times 10^{-5}}{1.152 \times 10^{-6}}$

$\simeq 14.1$s

- So stated threshold would be reached after about 14s

b) Radio tuning circuit

- Inductance of solenoid

$N = 600$, $l = 22.7$cm $= 0.227$m, $r = 1.42$cm $= 0.0142$m

$A = \pi r^2$

$= \pi (0.0142)^2$

$\simeq 6.33 \times 10^{-4}$ m^2

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$L = \frac{N\Phi}{I}$ $\Phi = BA$ $B = \mu_0 n I$ n = no. of turns per metre $\Phi = \frac{\mu_0 n I A \cdot N}{I}$

$L = \frac{\mu_0 N^2 A}{L}$

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For Solenoid;

$$L = \frac{\mu_0 N^2 A}{L}$$

$\mu_0 = 4\pi \times 10^{-7} Hm^{-1}$

$= \frac{4\pi \times 10^{-7} \times (600)^2 \times 6.33 \times 10^{-4}}{0.0227}$

$L \simeq 1.26 \times 10^{-3} H$

$\simeq 1.26 mH$

– Required capacitor for 2.516 MHz.

Resonance occurs when;

$$f = \frac{1}{2\pi\sqrt{LC}}$$

$$C = \frac{1}{(2\pi f)^2 L}$$

$f = 2.516 \times 10^6 Hz$, $L = 1.26 \times 10^{-3} H$

$$C = \frac{1}{(2\pi \times 2.516 \times 10^6)^2 \times (1.26 \times 10^{-3})}$$

$C \simeq 3.17 \times 10^{-12} F$

$\simeq 3.17 pF$

– Galvanometer doesnot deflect because the circuit is being supplied with $f = 40Hz$, but it has been tuned for 2.516 MHz. These two frequencies are different since at 40Hz, capacitor's reactance is extremely large.

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Impedance becomes resistive when resistance, R, is much greater than the reactance.

$Z = \sqrt{R^2 + X_L^2}$

$R \gg X_L \rightarrow Z \approx R$

Circuit behaves like an ordinary resistor.

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- Impedance - total opposition to flow of an AC in circuit having ~~both~~ resistors, capacitors and inductor.

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$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 40 \times 3.17 \times 10^{-12}} = 1.26 \times 10^9 \Omega$

$I = \frac{12}{1.26 \times 10^9}$

$= 9.5 \times 10^{-9} A$

$\simeq 9.5 nA$, current is extremely small.

- Expected galvanometer deflection

$I = \frac{k\theta}{NBA}$

$\theta = \frac{NBA}{k} I$

$= \frac{500 \times 0.25 \times 4.0 \times 10^{-4} \times 9.5 \times 10^{-9}}{3.0 \times 10^{-5}}$

$\simeq 1.6 \times 10^{-5}$ rad (which is very small and essentially zero on the galvanometer scale).

- How circuit should be modified?

- For the circuit to resonate with the incoming radio signal.

$f_{source} = 2.516 MHz$ and the capacitor should be approximately 3.17pF. Such that at resonance $X_L = X_C$ and impedance becomes mainly resistive allowing much large current.

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